

Appendix: Differentially Private Mechanisms using Enhanced Direction-Oriented Smooth Sensitivity

In this document, we provided the proofs of theorems and lemmas in the main paper.

Theorem 1. For all $i \in [m]$, $\tilde{S}_{f,\beta}^{*i+}$ and $\tilde{S}_{f,\beta}^{*i-}$ are enhanced direction-oriented β -smooth upper bound on LS_f^{i+} and LS_f^{i-} . In addition, for all $x \in D^n$, $\tilde{S}_{f,\beta}^{*i\pm}(x) \leq \tilde{S}^{i\pm}(x)$ holds for every enhanced direction-oriented β -smooth upper bound $\tilde{S}^{i+}$ and $\tilde{S}^{i-}$ on LS_f^{i+} and LS_f^{i-} .

Proof. From the definition, $\forall x \in D^n : \tilde{S}_{f,\beta}^{*i\pm}(x) \geq LS_f^{i\pm}(x)$ holds. We then show that $\forall x, y \in D^n, d(x, y) = 1$: $\tilde{S}_{f,\beta}^{*i\pm}(x) \leq e^\beta \cdot \tilde{S}_{f,\beta}^{*i\pm}(y) \wedge \tilde{S}_{f,\beta}^{*i\pm}(x) \leq e^\beta \cdot \tilde{S}_{f,\beta}^{*i\mp}(y)$.

(i) When $\tilde{S}_{f,\beta}^{*i+}(x) = LS_f^{i+}(x)$:

$$\begin{aligned} \tilde{S}_{f,\beta}^{*i\pm}(y) &\geq LS_f^{i+}(x) \cdot e^{-\beta \cdot d(x,y)} = e^{-\beta} \cdot LS_f^{i+}(x) \\ &= e^{-\beta} \cdot \tilde{S}_{f,\beta}^{*i+}(x). \end{aligned}$$

(ii) When $\tilde{S}_{f,\beta}^{*i+}(x) = LS_f^{i-}(x) \cdot e^{-2\beta}$:

$$\tilde{S}_{f,\beta}^{*i\pm}(y) \geq LS_f^{i-}(x) \cdot e^{-\beta} > \tilde{S}_{f,\beta}^{*i+}(x) > e^{-\beta} \cdot \tilde{S}_{f,\beta}^{*i+}(x).$$

(iii) When $\tilde{S}_{f,\beta}^{*i+}(x) = \max(LS_f^{i+}(x'), LS_f^{i-}(x')) \cdot e^{-\beta \cdot d(x',x)}$ for some $x' (\neq y) \in D^n$:

$$\begin{aligned} \tilde{S}_{f,\beta}^{*i\pm}(y) &\geq \max(LS_f^{i+}(x'), LS_f^{i-}(x')) \cdot e^{-\beta \cdot d(x',y)} \\ &\geq \max(LS_f^{i+}(x'), LS_f^{i-}(x')) \cdot e^{-\beta \cdot (d(x',x)+1)} \\ &[\because d(x',y) \leq d(x',x) + d(x,y)] \\ &= e^\beta \cdot \max(LS_f^{i+}(x'), LS_f^{i-}(x')) \cdot e^{-\beta \cdot d(x',x)} \\ &= e^{-\beta} \cdot \tilde{S}_{f,\beta}^{*i+}(x). \end{aligned}$$

(iv) When $\tilde{S}_{f,\beta}^{*i+}(x) = \max(LS_f^{i+}(y), LS_f^{i-}(y)) \cdot e^{-\beta \cdot d(y,x)}$:

(a) When $LS_f^{i+}(y) \geq LS_f^{i-}(y)$:

$$\begin{aligned} \tilde{S}_{f,\beta}^{*i+}(y) &\geq LS_f^{i+}(y) > LS_f^{i+}(y) \cdot e^{-\beta} \\ &= \tilde{S}_{f,\beta}^{*i+}(x) > e^{-\beta} \cdot \tilde{S}_{f,\beta}^{*i+}(x), \end{aligned}$$

$$\tilde{S}_{f,\beta}^{*i-}(y) \geq LS_f^{i+}(y) \cdot e^{-2\beta} = e^{-\beta} \cdot \tilde{S}_{f,\beta}^{*i+}(x).$$

(b) When $LS_f^{i+}(y) < LS_f^{i-}(y)$:

$$\text{As in case (a), } \tilde{S}_{f,\beta}^{*i\pm}(y) \geq e^{-\beta} \cdot \tilde{S}_{f,\beta}^{*i+}(x).$$

From (i) to (iv), $\forall x, y \in D^n, d(x, y) = 1$: $\tilde{S}_{f,\beta}^{*i+}(x) \leq e^\beta \cdot \tilde{S}_{f,\beta}^{*i\pm}(y)$ holds. In a similar manner, we can show that $\forall x, y \in D^n, d(x, y) = 1$: $\tilde{S}_{f,\beta}^{*i-}(x) \leq e^\beta \cdot \tilde{S}_{f,\beta}^{*i\pm}(y)$ holds.

Thereafter, we show that for all $x \in D^n$, $\tilde{S}_{f,\beta}^{*i\pm}(x) \leq \tilde{S}^{i\pm}(x)$. From the definition,

$$\tilde{S}^{i+}(x) \geq LS_f^{i+}(x),$$

$\forall y \in D^n, d(x, y) = 1$:

$$\begin{aligned} \tilde{S}^{i+}(x) &\geq e^{-\beta} \cdot \tilde{S}^{i\pm}(y) \geq e^{-2\beta} \cdot \tilde{S}^{i-}(x) \geq LS_f^{i-}(x) \cdot e^{-2\beta} \\ \wedge \tilde{S}^{i+}(x) &\geq e^{-\beta} \cdot \tilde{S}^{i\pm}(y) \geq e^{-\beta} \cdot LS_f^{i\pm}(y). \end{aligned}$$

Here, we assume that $\forall x', y \in D^n, d(x', y) = k : \tilde{S}^{i+}(x') \geq \max(LS_f^{i+}(y), LS_f^{i-}(y)) \cdot e^{-\beta \cdot d(y,x')}$ for some $k (> 0)$. Because for any $x, y \in D^n$ s.t. $d(x, y) = k+1, \exists x'$ satisfying $d(x, x') = 1 \wedge d(x', y) = k$,

$$\begin{aligned} \tilde{S}^{i+}(x) &\geq e^{-\beta} \cdot \tilde{S}^{i+}(x') \\ &\geq e^{-\beta} \cdot \max(LS_f^{i+}(y), LS_f^{i-}(y)) \cdot e^{-\beta \cdot d(y,x')} \\ &= \max(LS_f^{i+}(y), LS_f^{i-}(y)) \cdot e^{-\beta \cdot (d(y,x')+1)} \\ &= \max(LS_f^{i+}(y), LS_f^{i-}(y)) \cdot e^{-\beta \cdot d(y,x)}. \end{aligned}$$

Consequently,

$$\begin{aligned} \forall x, y \in D^n, x \neq y : \tilde{S}^{i+}(x) &\geq LS_f^{i+}(x) \\ \wedge \tilde{S}^{i+}(x) &\geq LS_f^{i-}(x) \cdot e^{-2\beta} \\ \wedge \tilde{S}^{i+}(x) &\geq \max(LS_f^{i+}(y), LS_f^{i-}(y)) \cdot e^{-\beta \cdot d(y,x)} \end{aligned}$$

holds, and $\forall x \in D^n : \tilde{S}^{i+}(x) \geq \tilde{S}_{f,\beta}^{*i+}(x)$ is derived accordingly. In a similar manner, we can show that $\forall x \in D^n : \tilde{S}^{i-}(x) \geq \tilde{S}_{f,\beta}^{*i-}(x)$ holds. $\square$

Theorem 2. For all $i \in [m]$, $\hat{S}_{f,\beta}^{*i+}$ and $\hat{S}_{f,\beta}^{*i-}$ are enhanced direction-oriented β -smooth upper bound on $\hat{LS}_f^{i+}$ and $\hat{LS}_f^{i-}$. In addition, for all $x \in D^n$, $\hat{S}_{f,\beta}^{*i\pm}(x) \leq \hat{S}^{i\pm}(x)$ holds for every enhanced direction-oriented β -smooth upper bound $\hat{S}^{i+}$ and $\hat{S}^{i-}$ on $\hat{LS}_f^{i+}$ and $\hat{LS}_f^{i-}$.

Proof. The proof is very similar to that of Theorem 1. $\square$

Theorem 3. For all $x \in D^n$ and $i \in [m]$, $\hat{S}_{f,\beta}^{*i+}(x) \leq \tilde{S}_{f,\beta}^{*i+}(x) \leq S_{f,\beta}^{*i+}(x) \wedge \hat{S}_{f,\beta}^{*i-}(x) \leq \tilde{S}_{f,\beta}^{*i-}(x) \leq S_{f,\beta}^{*i-}(x)$ holds.

Proof. Note that $\forall x \in D^n : \hat{LS}_f^{i+}(x) \leq LS_f^{i+}(x) \leq LS_f(x) \wedge \hat{LS}_f^{i-}(x) \leq LS_f^{i-}(x) \leq LS_f(x)$ holds from the definition and discussion in Section IV.A. Therefore,

$$\begin{aligned}
\hat{S}_{f,\beta}^{*i+}(x) &= \max \left(\hat{L}S_f^{i+}(x), \hat{L}S_f^{i-}(x) \cdot e^{-2\beta}, \right. \\
&\quad \left. \max_{y \in D^n, y \neq x} \left(\max \left(\hat{L}S_f^{i+}(y), \hat{L}S_f^{i-}(y) \right) \cdot e^{-\beta \cdot d(y,x)} \right) \right) \\
&\leq \max \left(LS_f^{i+}(x), LS_f^{i-}(x) \cdot e^{-2\beta}, \right. \\
&\quad \left. \max_{y \in D^n, y \neq x} \left(\max \left(LS_f^{i+}(y), LS_f^{i-}(y) \right) \cdot e^{-\beta \cdot d(y,x)} \right) \right) \\
&= \left(\tilde{S}_{f,\beta}^{*i+}(x) \right) \\
&\leq \max \left(LS_f(x), \max_{y \in D^n, y \neq x} \left(LS_f(y) \cdot e^{-\beta \cdot d(y,x)} \right) \right) \\
&= \max_{y \in D^n} \left(LS_f(y) \cdot e^{-\beta \cdot d(y,x)} \right) = S_{f,\beta}^*(x).
\end{aligned}$$

$\hat{S}_{f,\beta}^{*i-}(x) \leq \tilde{S}_{f,\beta}^{*i-}(x) \leq S_{f,\beta}^*(x)$ holds true in a similar manner. $\square$

Lemma 3. (ϵ -differentially private mechanism using an enhanced direction-oriented smooth upper bound based on improved DOLS)

Let h be an (α, β) -admissible noise probability density function on $\mathbb{R}$, where α and β are linear functions with respect to the input and $\alpha(0) = \beta(0) = 0$. Define h^+ and h^- as follows:

$$\begin{aligned}
h^+(z) &= \begin{cases} 2 \cdot h(z) & (z \geq 0) \\ 0 & (z < 0) \end{cases} \\
\text{and } h^-(z) &= \begin{cases} 0 & (z \geq 0) \\ 2 \cdot h(z) & (z < 0) \end{cases}.
\end{aligned}$$

Let z_i^+ and z_i^- be random variables derived from h^+ and h^- , respectively, for each $i \in [m]$ independently. Set $\alpha = \alpha(\epsilon)$ and $\beta = \beta(\frac{\epsilon}{m})$. For a function $f : D^n \rightarrow \mathbb{R}^m$, let $\hat{S}^{i+}, \hat{S}^{i-} : D^n \rightarrow \mathbb{R}$ be enhanced direction-oriented β -smooth upper bound on $\hat{L}S_f^{i+}$ and $\hat{L}S_f^{i-}$ for each $i \in [m]$. Thereafter, the mechanism M such that

$$\begin{aligned}
\forall i : M(x)_i &= f(x)_i + \begin{cases} \frac{\hat{S}^{i+}(x)}{\alpha} \cdot z_i^+ & (\text{with a probability of } 1/2) \\ \frac{\hat{S}^{i-}(x)}{\alpha} \cdot z_i^- & (\text{with a probability of } 1/2) \end{cases}
\end{aligned}$$

is ϵ -differentially private.

Proof. It is sufficient to show that for all x, y s.t. $d(x, y) = 1$,

$$\forall T \in \mathbb{R}^m : \Pr[M(x) = T] \leq e^\epsilon \cdot \Pr[M(y) = T].$$

Here, $\Pr[M(x) = T] = \prod_{i \in [m]} \Pr[M(x)_i = T_i]$, and

$$\Pr[M(x)_i = T_i] = \begin{cases} \frac{1}{2} \cdot \Pr \left[z_i^+ = \frac{T_i - f(x)_i}{N^{i+}(x)} \right] & (T_i \geq f(x)_i) \\ \frac{1}{2} \cdot \Pr \left[z_i^- = \frac{T_i - f(x)_i}{N^{i-}(x)} \right] & (T_i < f(x)_i) \end{cases},$$

where $N^{i\pm}(x) := \frac{\hat{S}^{i\pm}(x)}{\alpha}$.

(I) When $f(x)_i \leq f(y)_i$:

(i) When $T_i < f(x)_i$:

$$\begin{aligned}
&\Pr[M(x)_i = T_i] \\
&= \frac{1}{2} \cdot \Pr \left[z_i^- = \frac{T_i - f(x)_i}{N^{i-}(x)} \right] \\
&\leq \frac{1}{2} \cdot e^{\frac{\epsilon}{2m}} \cdot e^{\frac{f(y)_i - f(x)_i}{2 \cdot \hat{S}^{i-}(y)} \cdot \epsilon} \cdot \Pr \left[z_i^- = \frac{T_i - f(y)_i}{N^{i-}(y)} \right] \\
&\quad \left[\because \frac{N^{i-}(x)}{N^{i-}(y)} \leq e^{\beta(\frac{\epsilon}{m})}, \right. \\
&\quad \left. \frac{f(x)_i - f(y)_i}{N^{i-}(y)} = -\alpha \cdot \left(\frac{f(y)_i - f(x)_i}{\hat{S}^{i-}(y)} \cdot \epsilon \right) \right] \\
&= e^{\frac{\epsilon}{2m}} \cdot e^{\frac{f(y)_i - f(x)_i}{2 \cdot \hat{S}^{i-}(y)} \cdot \epsilon} \cdot \Pr[M(y)_i = T_i].
\end{aligned}$$

(ii) When $f(x)_i \leq T_i < f(y)_i$:

$$\begin{aligned}
&\Pr[M(x)_i = T_i] \\
&= \frac{1}{2} \cdot \Pr \left[z_i^+ = \frac{T_i - f(x)_i}{N^{i+}(x)} \right] \\
&\leq \frac{1}{2} \cdot e^{\frac{\epsilon}{2m}} \cdot e^{\frac{f(y)_i - f(x)_i}{2 \cdot \hat{S}^{i-}(y)} \cdot \epsilon} \cdot \Pr \left[z_i^+ = \frac{f(y)_i - T_i}{N^{i-}(y)} \right] \\
&\quad \left[\because \frac{N^{i+}(x)}{N^{i-}(y)} \leq e^{\beta(\frac{\epsilon}{m})}, \right. \\
&\quad \left. \frac{f(x)_i + f(y)_i - 2T_i}{N^{i-}(y)} \leq \alpha \left(\frac{f(y)_i - f(x)_i}{\hat{S}^{i-}(y)} \cdot \epsilon \right), \right. \\
&\quad \left. \frac{f(x)_i + f(y)_i - 2T_i}{N^{i-}(y)} > -\alpha \left(\frac{f(y)_i - f(x)_i}{\hat{S}^{i-}(y)} \cdot \epsilon \right) \right] \\
&= \frac{1}{2} \cdot e^{\frac{\epsilon}{2m}} \cdot e^{\frac{f(y)_i - f(x)_i}{2 \cdot \hat{S}^{i-}(y)} \cdot \epsilon} \cdot \Pr \left[z_i^- = \frac{T_i - f(y)_i}{N^{i-}(y)} \right] \\
&= e^{\frac{\epsilon}{2m}} \cdot e^{\frac{f(y)_i - f(x)_i}{2 \cdot \hat{S}^{i-}(y)} \cdot \epsilon} \cdot \Pr[M(y)_i = T_i].
\end{aligned}$$

(iii) When $f(y)_i \leq T_i$:

$$\begin{aligned}
\Pr[M(x)_i = T_i] &= \frac{1}{2} \cdot \Pr \left[z_i^+ = \frac{T_i - f(x)_i}{N^{i+}(x)} \right] \\
&\leq \frac{1}{2} \cdot \Pr \left[z_i^+ = \frac{T_i - f(y)_i}{N^{i+}(x)} \right] \\
&\quad [\because 0 \leq T_i - f(y)_i < T_i - f(x)_i] \\
&\leq \frac{1}{2} \cdot e^{\frac{\epsilon}{2m}} \cdot \Pr \left[z_i^+ = \frac{T_i - f(y)_i}{N^{i+}(y)} \right] \\
&\quad \left[\because \frac{N^{i+}(x)}{N^{i+}(y)} \leq e^{\beta(\frac{\epsilon}{m})} \right] \\
&= e^{\frac{\epsilon}{2m}} \cdot \Pr[M(y)_i = T_i].
\end{aligned}$$

Therefore, the following relation holds:

$$\Pr[M(x)_i = T_i] \leq e^{\frac{\epsilon}{2m}} \cdot e^{\frac{f(y)_i - f(x)_i}{2 \cdot \hat{S}^{i-}(y)} \cdot \epsilon} \cdot \Pr[M(y)_i = T_i].$$

(II) When $f(y)_i < f(x)_i$:

In a similar manner to (I), the following relation holds:

$$\Pr[M(x)_i = T_i] \leq e^{\frac{\epsilon}{2m}} \cdot e^{\frac{f(x)_i - f(y)_i}{2 \cdot \hat{S}^{i+}(y)} \cdot \epsilon} \cdot \Pr[M(y)_i = T_i].$$

From (I) and (II),

$$\begin{aligned} & \Pr[M(x)_i = T_i] \\ & \leq e^{\frac{\epsilon}{2m}} \cdot e^{\max\left(\frac{f(y)_i - f(x)_i}{2 \cdot \hat{S}^{i-}(y)}, \frac{f(x)_i - f(y)_i}{2 \cdot \hat{S}^{i+}(y)}\right) \cdot \epsilon} \cdot \Pr[M(y)_i = T_i] \end{aligned}$$

holds. Here, from the characteristic of the improved DOLS,

$$\begin{aligned} & \sum_{i \in [m]} \max \left(\frac{f(y)_i - f(x)_i}{2 \cdot \hat{S}^{i-}(y)}, \frac{f(x)_i - f(y)_i}{2 \cdot \hat{S}^{i+}(y)} \right) \\ & \leq \sum_{i \in [m]} \max \left(\frac{\max_{j \in [n]} c_j^{i-}(y)}{2 \cdot \hat{L}S_f^{i-}(y)}, \frac{\max_{j \in [n]} c_j^{i+}(y)}{2 \cdot \hat{L}S_f^{i+}(y)} \right) \leq \frac{1}{2}. \end{aligned}$$

Consequently, we can show that

$$\begin{aligned} \Pr[M(x) = T] &= \prod_{i \in [m]} \Pr[M(x)_i = T_i] \\ &\leq e^{\frac{\epsilon}{2}} \cdot e^{\frac{\epsilon}{2}} \cdot \prod_{i \in [m]} \Pr[M(y)_i = T_i] \\ &= e^{\epsilon} \cdot \Pr[M(y) = T]. \end{aligned}$$

Theorem 4. For all $\beta > 0$ and $i \in [m]$,

$$\begin{aligned} \hat{S}_{f,\beta}^{*i\pm}(x) &= \max \left(\hat{L}S_f^{i\pm}(x), \hat{L}S_f^{i\mp}(x) \cdot e^{-2\beta}, \right. \\ & \quad \left. \max_{y \in D^n, 1 \leq d(y,x) \leq \frac{1}{\beta} \cdot \ln \left(\frac{GS_f}{\max(\hat{L}S_f^{i+}(x), \hat{L}S_f^{i-}(x))} \right) + 2} \right. \\ & \quad \left. \left(\max(\hat{L}S_f^{i+}(y), \hat{L}S_f^{i-}(y)) \cdot e^{-\beta \cdot d(y,x)} \right) \right). \quad (1) \end{aligned}$$

Proof. When $d(y, x) > \frac{1}{\beta} \cdot \ln \left(\frac{GS_f}{\max(\hat{L}S_f^{i+}(x), \hat{L}S_f^{i-}(x))} \right) + 2$,

$$e^{-\beta \cdot d(y,x)} < \frac{\max(\hat{L}S_f^{i+}(x), \hat{L}S_f^{i-}(x))}{GS_f} \cdot e^{-2\beta}.$$

From this and that $\max(\hat{L}S_f^{i+}(y), \hat{L}S_f^{i-}(y)) \leq GS_f$,

$$\begin{aligned} & \max(\hat{L}S_f^{i+}(y), \hat{L}S_f^{i-}(y)) \cdot e^{-\beta \cdot d(y,x)} \\ & < \max(\hat{L}S_f^{i+}(x), \hat{L}S_f^{i-}(x)) \cdot e^{-2\beta} \\ & \leq \max(\hat{L}S_f^{i\pm}(x), \hat{L}S_f^{i\mp}(x) \cdot e^{-2\beta}). \end{aligned}$$

Consequently, from Definition 13, the relation (1) holds. $\square$

Theorem 5. We denote $K(x) := \max(\hat{L}S_f^{i+}(x), \hat{L}S_f^{i-}(x))$. For all β satisfying

$$\forall i \in [m]: e^{\beta} \geq \max_{x, x': d(x, x')=1} \frac{K(x')}{K(x)}, \quad (2)$$

the following equality holds for all $x \in D^n$:

$$\max_{y: y \neq x} (K(y) \cdot e^{-\beta \cdot d(y,x)}) = \max_{y: d(y,x)=1} (K(y) \cdot e^{-\beta}). \quad (3)$$

Furthermore, when

$$e^{\beta} \geq \max \left(\frac{\hat{L}S_f^{i\mp}(x)}{\hat{L}S_f^{i\pm}(x)}, 1 \right) \cdot \max_{x, x': d(x, x')=1} \frac{K(x')}{K(x)}, \quad (4)$$

$$\hat{S}_{f,\beta}^{*i\pm}(x) = \hat{L}S_f^{i\pm}(x).$$

Proof. We first show that when (2) holds,

$$\max_{y, d(y,x)=1} (K(y) \cdot e^{-\beta}) \geq K(y') \cdot e^{-\beta \cdot d(y',x)}$$

for all y' s.t. $d(y', x) > 1$.

(I) When $d(y', x) = 2$, there exists some y'' satisfying $d(y', y'') = 1 \wedge d(y'', x) = 1$. Therefore,

$$\begin{aligned} \max_{y, d(y,x)=1} (K(y) \cdot e^{-\beta}) &\geq K(y'') \cdot e^{-\beta} \\ &\geq (K(y') \cdot e^{-\beta}) \cdot e^{-\beta} \quad [\because (2)] \\ &= K(y') \cdot e^{-\beta \cdot d(y',x)}. \end{aligned}$$

(II) We assume that $\max_{y, d(y,x)=1} (K(y) \cdot e^{-\beta}) \geq K(y'') \cdot e^{-\beta \cdot d(y'',x)}$ holds for all y'' satisfying $d(y'', x) = k$. Here, for any y' s.t. $d(y', x) > k$, there exists some y'' satisfying $d(y', y'') = 1 \wedge d(y'', x) = k$. Therefore, as in case (I),

$$\max_{y, d(y,x)=1} (K(y) \cdot e^{-\beta}) \geq K(y') \cdot e^{-\beta \cdot d(y',x)}.$$

From (I) and (II), the relation (3) holds for all x .

Furthermore, when (4) holds,

$$\begin{aligned} \max_{y, d(y,x)=1} (K(y) \cdot e^{-\beta}) &\leq K(x) \cdot \frac{1}{\max \left(\frac{\hat{L}S_f^{i\mp}(x)}{\hat{L}S_f^{i\pm}(x)}, 1 \right)} \\ &= \hat{L}S_f^{i\pm}(x). \end{aligned}$$

In addition, $\hat{L}S_f^{i\pm}(x) \geq \hat{L}S_f^{i\mp}(x) \cdot e^{-2\beta}$ holds. Consequently, $\hat{S}_{f,\beta}^{*i\pm}(x) = \hat{L}S_f^{i\pm}(x)$. $\square$

Lemma 4. (Unbiased ϵ -differentially private mechanism using an enhanced direction-oriented smooth upper bound based on improved DOLS)

Let h be a symmetric (α, β) -admissible noise probability density function on $\mathbb{R}$, where α and β are linear functions with respect to the input and $\alpha(0) = \beta(0) = 0$. Define h^+ and h^- as follows:

$$\begin{aligned} h^+(z) &= \begin{cases} 2 \cdot h(z) & (z \geq 0) \\ 0 & (z < 0) \end{cases} \\ \text{and } h^-(z) &= \begin{cases} 0 & (z \geq 0) \\ 2 \cdot h(z) & (z < 0) \end{cases}. \end{aligned}$$

Let z_i^+ and z_i^- be random variables derived from h^+ and h^- , respectively, for each $i \in [m]$ independently. Set $\alpha = \alpha(\epsilon)$ and $\beta = \beta(\frac{\epsilon}{m})$. For a function $f : D^n \rightarrow \mathbb{R}^m$, let $\hat{S}^{i+}, \hat{S}^{i-} : D^n \rightarrow \mathbb{R}$ be enhanced direction-oriented β -smooth upper bound on $\hat{L}S_f^{i+}$ and $\hat{L}S_f^{i-}$ for each $i \in [m]$. Thereafter, the mechanism M such that

$$\forall i: M(x)_i = f(x)_i \\ + \left\{ \frac{\hat{S}^{i+}(x)}{\alpha} \cdot z_i^+ \quad \left(\text{with a probability of } \frac{\hat{S}^{i-}(x)}{\hat{S}^{i+}(x) + \hat{S}^{i-}(x)} \right) \right. \\ \left. \frac{\hat{S}^{i-}(x)}{\alpha} \cdot z_i^- \quad \left(\text{with a probability of } \frac{\hat{S}^{i+}(x)}{\hat{S}^{i+}(x) + \hat{S}^{i-}(x)} \right) \right\}$$

is $(\epsilon + 2m\beta)$ -differentially private.

Proof. It is sufficient to show that for all x, y s.t. $d(x, y) = 1$,

$$\forall T \in \mathbb{R}^m: \Pr[M(x) = T] \leq e^{\epsilon + 2m\beta} \cdot \Pr[M(y) = T].$$

Here, $\Pr[M(x) = T] = \prod_{i \in [m]} \Pr[M(x)_i = T_i]$, and

$$\Pr[M(x)_i = T_i] \\ = \begin{cases} \frac{\hat{S}^{i-}(x)}{\hat{S}^{i+}(x) + \hat{S}^{i-}(x)} \cdot \Pr\left[z_i^+ = \frac{T_i - f(x)_i}{N^{i+}(x)}\right] & (T_i \geq f(x)_i) \\ \frac{\hat{S}^{i+}(x)}{\hat{S}^{i+}(x) + \hat{S}^{i-}(x)} \cdot \Pr\left[z_i^- = \frac{T_i - f(x)_i}{N^{i-}(x)}\right] & (T_i < f(x)_i) \end{cases},$$

where $N^{i\pm}(x) := \frac{\hat{S}^{i\pm}(x)}{\alpha}$.

(I) When $f(x)_i \leq f(y)_i$:

(i) When $T_i < f(x)_i$:

$$\frac{\hat{S}^{i+}(x) + \hat{S}^{i-}(x)}{\hat{S}^{i+}(x)} \cdot \Pr[M(x)_i = T_i] \\ = \Pr\left[z_i^- = \frac{T_i - f(x)_i}{N^{i-}(x)}\right] \\ \leq e^{\frac{\epsilon}{2m}} \cdot e^{\frac{f(y)_i - f(x)_i}{2 \cdot \hat{S}^{i-}(y)} \cdot \epsilon} \cdot \Pr\left[z_i^- = \frac{T_i - f(y)_i}{N^{i-}(y)}\right] \\ = \frac{\hat{S}^{i+}(y) + \hat{S}^{i-}(y)}{\hat{S}^{i+}(y)} \cdot e^{\frac{\epsilon}{2m}} \cdot e^{\frac{f(y)_i - f(x)_i}{2 \cdot \hat{S}^{i-}(y)} \cdot \epsilon} \cdot \Pr[M(y)_i = T_i].$$

(ii) When $f(x)_i \leq T_i < f(y)_i$:

$$\frac{\hat{S}^{i+}(x) + \hat{S}^{i-}(x)}{\hat{S}^{i-}(x)} \cdot \Pr[M(x)_i = T_i] \\ = \Pr\left[z_i^+ = \frac{T_i - f(x)_i}{N^{i+}(x)}\right] \\ \leq e^{\frac{\epsilon}{2m}} \cdot e^{\frac{f(y)_i - f(x)_i}{2 \cdot \hat{S}^{i+}(y)} \cdot \epsilon} \cdot \Pr\left[z_i^+ = \frac{f(y)_i - T_i}{N^{i+}(y)}\right] \\ = e^{\frac{\epsilon}{2m}} \cdot e^{\frac{f(y)_i - f(x)_i}{2 \cdot \hat{S}^{i+}(y)} \cdot \epsilon} \cdot \Pr\left[z_i^- = \frac{T_i - f(y)_i}{N^{i-}(y)}\right] \\ = \frac{\hat{S}^{i+}(y) + \hat{S}^{i-}(y)}{\hat{S}^{i+}(y)} \cdot e^{\frac{\epsilon}{2m}} \cdot e^{\frac{f(y)_i - f(x)_i}{2 \cdot \hat{S}^{i+}(y)} \cdot \epsilon} \cdot \Pr[M(y)_i = T_i].$$

(iii) When $f(y)_i \leq T_i$:

$$\frac{\hat{S}^{i+}(x) + \hat{S}^{i-}(x)}{\hat{S}^{i-}(x)} \cdot \Pr[M(x)_i = T_i] \\ = \Pr\left[z_i^+ = \frac{T_i - f(x)_i}{N^{i+}(x)}\right] \\ \leq \Pr\left[z_i^+ = \frac{T_i - f(y)_i}{N^{i+}(x)}\right] \\ \leq e^{\frac{\epsilon}{2m}} \cdot \Pr\left[z_i^+ = \frac{T_i - f(y)_i}{N^{i+}(y)}\right] \\ = \frac{\hat{S}^{i+}(y) + \hat{S}^{i-}(y)}{\hat{S}^{i-}(y)} \cdot e^{\frac{\epsilon}{2m}} \cdot \Pr[M(y)_i = T_i].$$

Here, the following inequalities hold:

$$\frac{\hat{S}^{i+}(y) + \hat{S}^{i-}(y)}{\hat{S}^{i+}(x) + \hat{S}^{i-}(x)} \leq e^\beta, \\ \frac{\hat{S}^{i+}(x)}{\hat{S}^{i+}(y)} \leq e^\beta, \quad \frac{\hat{S}^{i-}(x)}{\hat{S}^{i+}(y)} \leq e^\beta, \quad \frac{\hat{S}^{i-}(x)}{\hat{S}^{i-}(y)} \leq e^\beta.$$

Therefore, the following relation holds:

$$\Pr[M(x)_i = T_i] \\ \leq e^{2\beta} \cdot e^{\frac{\epsilon}{2m}} \cdot e^{\frac{f(y)_i - f(x)_i}{2 \cdot \hat{S}^{i-}(y)} \cdot \epsilon} \cdot \Pr[M(y)_i = T_i].$$

(II) When $f(y)_i < f(x)_i$:

In a similar manner to (I), the following relation holds:

$$\Pr[M(x)_i = T_i] \\ \leq e^{2\beta} \cdot e^{\frac{\epsilon}{2m}} \cdot e^{\frac{f(x)_i - f(y)_i}{2 \cdot \hat{S}^{i+}(y)} \cdot \epsilon} \cdot \Pr[M(y)_i = T_i].$$

From (I) and (II),

$$\Pr[M(x)_i = T_i] \\ \leq e^{\frac{\epsilon}{2m}} \cdot e^{\max\left(\frac{f(y)_i - f(x)_i}{2 \cdot \hat{S}^{i-}(y)}, \frac{f(x)_i - f(y)_i}{2 \cdot \hat{S}^{i+}(y)}\right) \cdot \epsilon} \cdot \Pr[M(y)_i = T_i]$$

holds. Here, from the characteristic of the improved DOLS,

$$\sum_{i \in [m]} \max\left(\frac{f(y)_i - f(x)_i}{2 \cdot \hat{S}^{i-}(y)}, \frac{f(x)_i - f(y)_i}{2 \cdot \hat{S}^{i+}(y)}\right) \\ \leq \sum_{i \in [m]} \max\left(\frac{\max_{j \in [n]} c_j^{i-}(y)}{2 \cdot \hat{L}S_f^{i-}(y)}, \frac{\max_{j \in [n]} c_j^{i+}(y)}{2 \cdot \hat{L}S_f^{i+}(y)}\right) \leq \frac{1}{2}.$$

Consequently, we can show that

$$\Pr[M(x) = T] = \prod_{i \in [m]} \Pr[M(x)_i = T_i] \\ \leq e^{2m\beta} \cdot e^{\frac{\epsilon}{2}} \cdot e^{\frac{\epsilon}{2}} \cdot \prod_{i \in [m]} \Pr[M(y)_i = T_i] \\ = e^{\epsilon + 2m\beta} \cdot \Pr[M(y) = T].$$

□

Theorem 6. Let M be the mechanism described in Lemma 4, and the following relation holds:

$$\forall x \in D^n: \forall i: \mathbb{E}[M(x)_i] = f(x)_i.$$

Proof. For any $x \in D^n$ and i ,

$$\mathbb{E}[M(x)_i] = f(x)_i + \frac{\hat{S}^{i-}(x)}{\hat{S}^{i+}(x) + \hat{S}^{i-}(x)} \cdot \frac{\hat{S}^{i+}(x)}{\alpha} \cdot \mathbb{E}[z_i^+] \\ + \frac{\hat{S}^{i+}(x)}{\hat{S}^{i+}(x) + \hat{S}^{i-}(x)} \cdot \frac{\hat{S}^{i-}(x)}{\alpha} \cdot \mathbb{E}[z_i^-] \\ = f(x)_i \cdot [\cdot \mathbb{E}[z_i^+] = -\mathbb{E}[z_i^-]]$$

□